

## ASSIGNMENT 1

1. Create the function PersonName which accepts three parameters (FirstName, LastName, Age).

Enhance it by adding a condition: if you are above 25, the code should display “[Your full name]. You are an adult person”, otherwise - “[Your full name]. You are a young person”.

Call your function PersonName.

Code:-

```
1  DELIMITER $$
2
3  CREATE FUNCTION PersonName(FirstName VARCHAR(50), LastName VARCHAR(50), Age INT)
4  RETURNS VARCHAR(200) DETERMINISTIC
5  BEGIN
6      DECLARE FullName VARCHAR(100);
7      DECLARE Message VARCHAR(200);
8
9      SET FullName = CONCAT(FirstName, ' ', LastName);
10
11     IF Age > 25 THEN
12         SET Message = CONCAT(FullName, '. You are an adult person');
13     ELSE
14         SET Message = CONCAT(FullName, '. You are a young person');
15     END IF;
16
17     RETURN Message;
18 END $$
19
20 DELIMITER ;
21
22 SELECT PersonName('Sajan', 'Singh', 25);
23 SELECT PersonName('Rahul', 'Dua', 30);
24
```

Output:-

```
Line 1  
[mysql> source /Users/sajanshergill/Desktop/assignment2.sql  
ERROR 1304 (42000): FUNCTION PersonName already exists  
+-----+  
| PersonName('Sajan', 'Singh', 25) |  
+-----+  
| Sajan Singh. You are a young person |  
+-----+  
1 row in set (0.00 sec)  
  
+-----+  
| PersonName('Rahul', 'Dua', 30) |  
+-----+  
| Rahul Dua. You are an adult person |  
+-----+  
1 row in set (0.00 sec)  
  
mysql> █
```

2. Create the function average5 which accepts 5 integer parameters and returns their average.

Call your function average5 with the parameters 1,2,3,4,5

Code:-

```
Users > sajanshergill > Desktop > assignment 2.sql  
1 DELIMITER $$  
2  
3 CREATE FUNCTION average5(Num1 INT, Num2 INT, Num3 INT, Num4 INT, Num5 INT)  
4 RETURNS FLOAT DETERMINISTIC  
5 BEGIN  
6 | RETURN (Num1 + Num2 + Num3 + Num4 + Num5) / 5.0;  
7 END $$  
8  
9 DELIMITER ;  
10  
11  
12 SELECT average5(1, 2, 3, 4, 5);  
13 |
```

Output:-

```
[mysql> source /Users/sajanshergill/Desktop/assignment 2.sql
Query OK, 0 rows affected (0.01 sec)

+-----+
| average5(1, 2, 3, 4, 5) |
+-----+
|                        3 |
+-----+
1 row in set (0.00 sec)

mysql>
```

3. Create the function varianceA which accepts a single numeric parameter (A) and calculates the variance of the numbers 1 to A.  
Hint: Probably, you will want to call the second function inside this one.  
Call the function with the parameter 5

Code:-

```
1  DELIMITER $$
2
3  CREATE FUNCTION varianceA(A INT)
4  RETURNS FLOAT DETERMINISTIC
5  BEGIN
6      DECLARE SumX FLOAT DEFAULT 0;
7      DECLARE SumSquaredDiff FLOAT DEFAULT 0;
8      DECLARE Mean FLOAT;
9      DECLARE Counter INT DEFAULT 1;
10
11      -- Calculate the Mean
12      WHILE Counter <= A DO
13          SET SumX = SumX + Counter;
14          SET Counter = Counter + 1;
15      END WHILE;
16
17      SET Mean = SumX / A;
```

```

17     SET Mean = SumX / A;
18
19     -- Reset Counter and Calculate Variance
20     SET Counter = 1;
21     SET SumX = 0;
22
23     WHILE Counter <= A DO
24         SET SumSquaredDiff = SumSquaredDiff + POW(Counter - Mean, 2);
25         SET Counter = Counter + 1;
26     END WHILE;
27
28     RETURN SumSquaredDiff / A;
29 END $$
30
31 DELIMITER ;
32
33
34
35 SELECT varianceA(5);
36

```

Output:-

```

1 row in set (0.00 sec)

[mysql> source /Users/sajanshergill/Desktop/assignment 2.3.sql
Query OK, 0 rows affected (0.01 sec)

+-----+
| varianceA(5) |
+-----+
|          2 |
+-----+
1 row in set (0.00 sec)

mysql> █

```